

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	SECTION:	CPE21S2
DATE PERFORMED:	07-08-26	DATE SUBMITTED:	07 - 10 - 26
NAME:	Paula Esguerra	INSTRUCTOR:	Engr. Jimlord Quejado

SEATWORK NO. 1.1 [Seatwork 1.1 Algorithm, Flowchart, Pseudocode]

1. PROCEDURE OUTPUT

What procedures did you perform?

The procedure that I did was to try to recall how to do pseudocode and flowchart in a better way too, then I started searching what algorithm is. So when at algorithm I went with the basics and after I finished them, I started to think in question 1 on how about I improve some points so I added a "while" loop, so I set the total to 0 and added a counter, the counter is for the quizzes one to three, so as the loop goes, the total starts at zero and plus the number given by the user, up until quiz number three. Then when it was done because I only programmed the code until less than or equal to three, it picked no in the decision, fig 1, after adding the three scores, it was divided by three to get the average, then it displayed total and average at the end of the code, fig 3. Furthermore, at question 2, I started at the basics then thought of adding "for" loop, it actually was a good idea, I searched for idea in gemini on how to do it, then I followed some of the instruction and recalled my way to finishing the code, so some of the terms there looks like ai generated, so then, I started to set the total again at zero, add the "for" loop after the conditions was as little bit similar to question 1, I put on a decision until it was satisfied it would repeat 3 times then go to the next decision where it will be subtracted to the payment of the input of the user, after that the total and the change will be displayed at the end but I didn't displayed the amount of the prices of the user input, I only displayed the change, the payment and the total..

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Q1. Compute the total and average of 3 quizzes then display the total and average.

Algorithm

Step 1: Start

Step 2: Set the Total to 0 and the Counter to 1

Step 3: While Counter is less than or equal to 3, repeat these steps: Input a quiz score; Add the quiz score to Total; Add 1 to Counter

Step 4: Divide the Total by 3 to find the Average

Step 5: Display Total and Average

Step 6: End

Pseudocode

Start

Total = 0

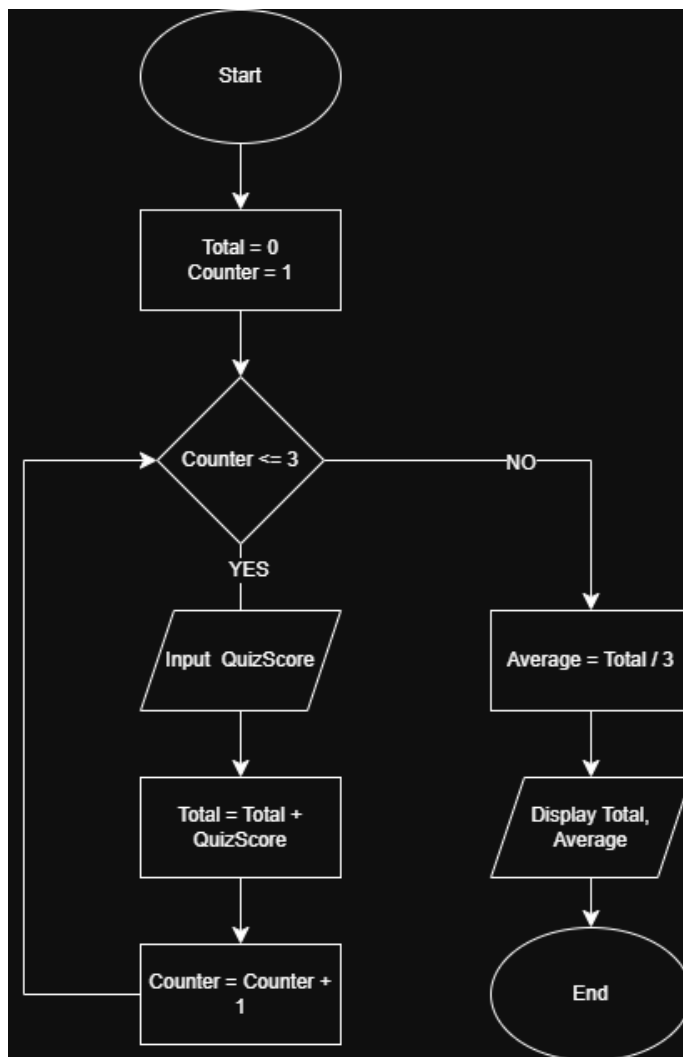
Counter = 1

```
While Counter <= 3 Do
    Input QuizScore
    Total = Total + QuizScore
    Counter = Counter + 1
EndWhile
```

Average = Total / 3

Display Total, Average

End



```

1  #include<iostream>
2
3
4  int main ( ) {
5      double Total = 0;
6      double QuizScore = 0;
7      int counter = 1;
8
9      while (counter <= 3) {
10         std::cout << "Input the score for Quiz " << counter << ":" << std::endl;
11         std::cin >> QuizScore;
12         Total = Total + QuizScore;
13         counter = counter + 1;
14     }
15
16     double average = Total / 3;
17
18     std::cout << "====Results====" << std::endl;
19     std::cout << "Total Score: " << Total << std::endl;
20
21     std::cout << "Average Score: " << average << std::endl;
22
23     return 0;
24 }
25
26
27

```

D:\CPE 010\ss.exe

```

Input the score for Quiz 1:
22
Input the score for Quiz 2:
67
Input the score for Quiz 3:
33
====Results====
Total Score: 122
Average Score: 40.6667

-----
Process exited after 5.266 seconds with return value 0
Press any key to continue . . . |

```

Q2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

Algorithm

Step 1: Start

Step 2: Set Total Price to 0

Step 3: Repeat the following steps 3 times: Input the price of an item; Add the item price to the Total Price

Step 4: Input the cash Payment from the customer

Step 5: Subtract the Total Price from the Payment

Step 6: Store the result as Change

Step 7: Display the Total Price, Payment, Change

Step 8: End

Pseudocode

Start

 Total = 0

 For i = 1 To 3 Do

 Input Price

 Total = Total + Price

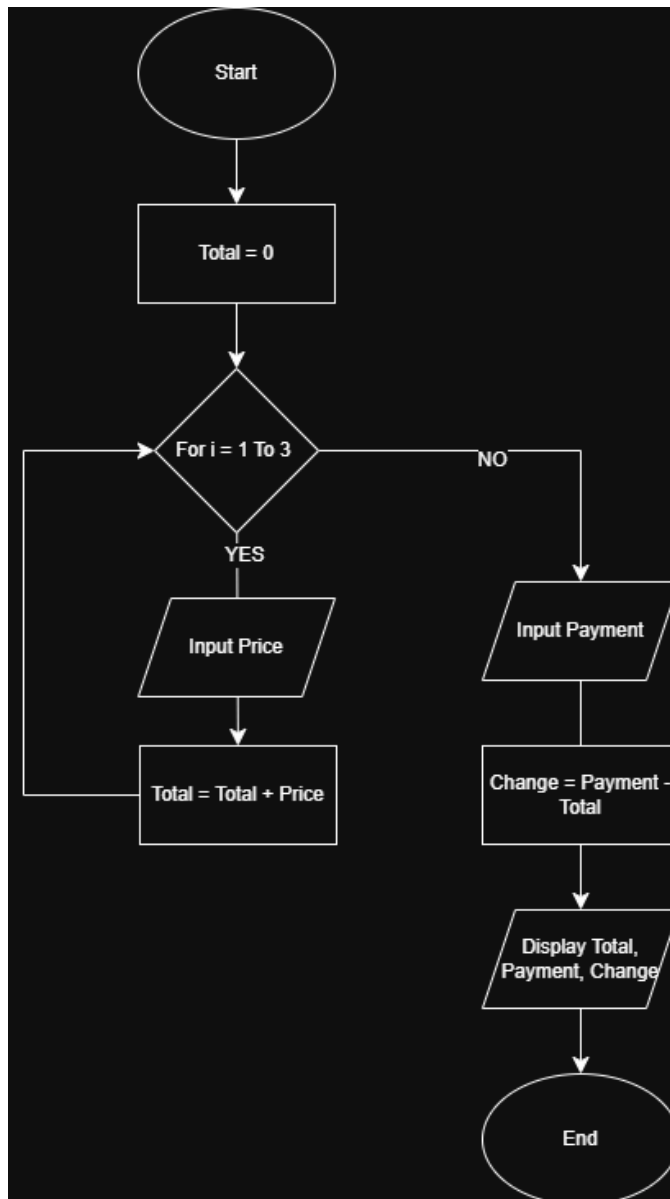
 EndFor

 Input Payment

 Change = Payment - Total

 Display Total, Payment, Change

End



```
1  #include<iostream>
2
3
4  int main ( ) {
5      double TotalPR = 0;
6      double price = 0;
7      double payment = 0;
8      double change = 0;
9
10     for (int i = 1; i <= 3; i++) {
11         std::cout << "Input the price for item " << i << ": " ;
12         std::cin >> price;
13         TotalPR = TotalPR + price;
14     }
15
16     std::cout << "\n Amount Total: " << TotalPR << std::endl;
17     std::cout << "Customer Cash Payment: ";
18     std::cin >> payment;
19
20     change = payment - TotalPR;
21
22     std::cout << "\n===Transaction Receipt===" <<std::endl;
23     std::cout << "Total Price: " << TotalPR << std::endl;
24     std::cout << "Payment: " << payment <<std::endl;
25     std::cout << "Change: " << change << std::endl;
26
27
28     return 0;
29
30 }
31
```

```
D:\CPE 010\ss.exe  X + v
Input the price for item 1:1
Input the price for item 2:3
Input the price for item 3:4

Amount Total: 8
Customer Cash Payment: 44

===Transaction Receipt===
Total Price: 8
Payment: 44
Change: 36

-----
Process exited after 6.052 seconds with return value 0
Press any key to continue . . .
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Idea generation, and Checking if my work is correct.

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.



COMPUTER
ENGINEERING

QUEZON CITY

Extent of Use: After I did my work, the algorithms, pseudocode, I copied and pasted in gemini to check if it is correct, and some are not too detailed enough, so as gemini advised that I change some terms, and so I did although the terms are different, my original idea is still embedded with I did..

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.